# Pulkit Kumar

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**Pre-final year electrical engineering student with experience in embedded systems across C/C++ firmware development, deterministic multi-threading, automotive electronics & edge AI perception/control stacks. 3+ provisional patents possessed across automotive electronics, edge-cloud data fusion, & digital twin modelling.**

## Technical Skills

**Languages**: MISRA C/C++, Rust, Python, Verilog, Java, Bash, MATLAB
**Embedded**: Yocto, Buildroot, CMake, UEFI, Linux Kernel, FreeRTOS, ROS2, Simulink, AUTOSAR
**Protocols**: CAN bus, Ethernet, RF, USART, LIN, I2C, SPI, I2S, WebSockets, TCP/IP
**AI/Data**: PyTorch, OpenCV, YOLO, TensorRT, TensorFlow, Keras, Prometheus, Grafana
**Tools**: GCC, GDB, Make, GNU toolchain, Git, GitLab, Coverity, Tracealyzer, Jenkins, Docker
**BSP/HAL**: STM32, NXP, Renesas, Nvidia Jetson, ESP32, Raspberry Pi

## Experience

**Electronics Head** — January 2024 – Present
*University Racing - Team SolarMobil* — *Manipal, IN*

- Built IS26262-aligned safety-critical embedded software for solar racing EVs on STM32 (Cortex M7 & M4) platforms with FreeRTOS, interfacing 40+ sensor drivers over CAN for custom ECUs and TouchGFX displays
- Optimized task scheduling and DMA-driven data transfer to significantly reduce UI latency, driver-side control response delays, and power consumption in race environments- contributing to 2 patent filings
- Prototyped closed-track autonomous navigation using YOLOv2 + ROS2 perception, LiDAR-based localization & MPC control, evaluated 20m sessions with stable performance under ∼30W on a power-constrained Jetson Orin

**System Engineer** — May 2025 – July 2025
*Internship - Symbionic* — *Chennai, IN*

- Developed drivers for ESP32 cores and designed a signal acquisition pipeline for multi-channel EMG/EEG sensing with preprocessing for spatiotemporal feature extraction (STFT)
- Deployed a multimodal edge AI pipeline using LSTM + transformer encoders on the Jetson platform; achieved stable sub-second control responsiveness with ∼95% classification accuracy during evaluation

**Digital Design Engineer** — May 2024 – July 2024
*Internship - CoreEL Technologies* — *Bengaluru, IN*

- Designed RTL modules for CAN and I2C controllers in Verilog based on microarchitecture specifications; collaborated with verification teams on functional validation workflows
- Brought designs onto a Zynq-7000 platform using Vivado/Vitis; worked through synthesis, constraint handling and debugging to validate hardware behavior on FPGAs

## Education

**Manipal Institute of Technology** — Manipal, IN
*Bachelor of Technology in Electrical & Electronics Engineering* — *Aug. 2023 – May 2027 (Expected)*

## Projects

**Interactive Hologram** | *ARM64 Linux, CSI, JetPack, I2S, ALSA, YOLO, TensorRT, CUDA, OpenCV* — Sept 2025

- Building a holographic computing system with sub-250ms multimodal real-time agentic control of Linux applications via Wayland and gesture-to-IPC mapping with live inference for CSI camera and I2S audio feeds
- Integrating a TI LightCrafter holographic display and a Leap motion controller on the Jetson AGX platform with SwayWM compositor, Riva/Whisper NLP, and TensorRT/VPI acceleration on a customized yocto image

**Patient Monitoring System** | *ECG, IMU, DSP, WebSockets, Influx, Grafana, OpenCV, Altium, ESP32* — Nov 2024

- Developed a comprehensive vitals monitoring system with custom DSP algorithms for ECG/GSR sensing and a novel gyroscope-based respiration metric, streaming via WebSockets into Influx + Grafana dashboards
- Enabled near real-time ward-level visualization; evaluation demonstrated built-in emergency alerts reducing response times to under 5 minutes